batch of data → loss L(θ)

backpropagation
gradient g

every optimizer
θ ← θ − η · u(g, state)

remember the past?
momentum · variance reduction

rescale each coordinate?
adaptive · per-parameter steps

use curvature or
matrix structure?
second-order · geometric

state too large?
memory-efficient · low-rank

tuning the schedule painful?
schedule-free · parameter-free

landscape too rugged?
evolutionary · population